Final Exam Solution - SOC2101: Soc., Tech. and Engg. Ethics

Q1: Analysis using the ACM Code of Ethics

The scenario involves Quinn, a member of a medical research team studying genetic factors and social behaviour. The tool built by Quinn links anonymized data sets but has a bug that causes some records to be linked incorrectly. Although the team was granted approval from the ethics review board (ERB) for the study, Quinn suggests releasing the source code to facilitate peer review.

Analysis of ACM Code of Ethics:
1. Public Interest (1.1): The ACM Code emphasizes the need to prioritize the public good, which aligns with Quinn's concern about the bug and the potential impact on the research. The team’s commitment to anonymizing data shows a commitment to public welfare.
2. Honesty (2.3): Quinn raised concerns about the bug and suggested releasing the source code. This aligns with the ACM principle of honesty, which urges professionals to avoid conflicts of interest and ensure the public is fully informed.
3. Fairness (1.7): The tool is designed to respect privacy, and releasing the source code would enhance fairness by allowing others to identify bugs and potential problems.
4. Privacy (1.6): The team followed privacy guidelines by anonymizing data, but the bug that caused erroneous linking may have compromised individual privacy, violating this principle. Ensuring the data is completely anonymous is crucial to adhering to the privacy requirement.
5. Quality (2.1): The bug and the potential errors in linking data indicate a failure to meet the quality standards expected. This highlights the importance of rigorous testing, especially when dealing with sensitive data.

Q2: Bias in Google Autocomplete

The Google search autocompletion example shows a historical bias. The suggestion that "Sri Lankans are terrorists" reflects prejudices ingrained in society and historical data used to train the algorithm.

Historical Bias: This bias occurs due to the data used to train the autocomplete system, which reflects past stereotypes or biased information present in historical sources or society.

Solution: Google can address this issue by:
1. Auditing the data used to train the algorithm to remove biased content.

2. Implementing bias correction techniques and making adjustments to ensure more neutral suggestions.
3. Allowing users to report biased autocomplete suggestions to refine the system further.

Q3: Ethical Measures for Twitter's Research

Twitter conducted a research project involving the manipulation of content seen by users without explicit consent. This raises ethical concerns about user privacy and informed consent.

Ethical Measures:
1. Informed Consent: Twitter should have obtained explicit consent from users before manipulating the content they see, explaining the nature of the study and its potential impact.
2. Transparency: Users should be made aware of how their data is being used, with clear information provided about the study's purpose and methodology.
3. Ethical Review: An independent ethics review board should have evaluated the study to ensure it adheres to ethical standards, particularly regarding consent and user autonomy.
4. Privacy: Ensuring user data is anonymized and protected is crucial in any research involving social media manipulation.

Q4: Ethical Challenges in Brain-Computer Interface (BCI) Study

The ethical dilemmas surrounding RDJ's BCI study are numerous:
1. Informed Consent: Participants might not fully understand the technology, making informed consent more complicated. There is a need for clear communication and possibly third-party assessment to ensure that participants fully comprehend the risks.
2. Privacy and Security: Accessing neural data presents significant privacy issues. Proper measures must be taken to protect sensitive data from unauthorized access and potential misuse.
3. Long-Term Effects and Consent Withdrawal: The potential for long-term neurological changes raises concerns about participants' ability to withdraw consent at a later stage without adverse consequences.
4. Safety: The invasive nature of BCI technology raises safety concerns. Strict safety protocols, including monitoring and emergency procedures, are necessary.
5. Societal Inequality: There is a risk of exacerbating inequalities by limiting access to the technology. RDJ's company should ensure equitable access to the technology for all demographics.

Q5: Consumer Opinion on Replaceable vs. Non-Removable Batteries

From a consumer perspective, the EU's directive for replaceable batteries makes sense for the following reasons:
1. Environmental Impact: Replaceable batteries reduce electronic waste by allowing consumers to replace batteries instead of discarding an entire device.
2. Cost-Effectiveness: Replacing the battery extends the lifespan of the phone, saving consumers money in the long term by avoiding the need to buy a new device.

However, the smartphone industry argues that non-removable batteries reduce manufacturing costs and make devices lighter. Moreover, phone performance may degrade over time, prompting consumers to replace the entire device, which could be seen as an opportunity for better technology.

Analysis:
While the industry's argument regarding cost and performance is valid, the environmental and consumer benefits of replaceable batteries outweigh the drawbacks. Consumers could benefit from longer device lifespans and reduced electronic waste.
There should be a balance, where manufacturers offer devices with replaceable batteries while maintaining other benefits like performance and design.